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# Online identification of laser welding penetration through multi-photoelectric decomposition-reconstruction and shifted-windows-based transformer deep learning framework



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## Outline

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## Highlights

- The time-dependent dynamics of the plasma plume and keyhole exhibit a strong correlation with the photoelectric signal.
- VMD-FEA algorithm can extract the signal components corresponding to the strongest correlation to penetration depth.
- The proposed model with shifted-windows-based transformer shows a high classification ability for predicting the weld penetration.

## Abstract

Measurements of optical radiation during laser deep-penetration welding are closely related to plasma plume, keyhole, and melt pool behaviors. Mining effective optical information and accurately assessing welding quality still pose significant challenges. Our research aims to capture and deeply analyze multi-photoelectric signals and build a novel framework for recognizing penetration states. A multi-source monitoring platform based on photoelectric sensors and high-speed photography was constructed to collect optical radiation, plasma plume, and melt pool images during pulsed laser welding of aluminum alloy. The relationship between the photoelectric signals and penetration depth was systematically investigated using Variational Modal Decomposition and Frequency-domain Energy Analysis (VMD-FEA) and visual representation. An efficient model with shifted-windows-based transformer blocks (STBs) outperforms CNN-based models and achieves 98.8% accuracy in identifying penetration states. Our work presents a new paradigm for the online monitoring of intelligent laser manufacturing, further advancing towards real-time process control.

## Keywords

Aluminum alloys laser welding; Photoelectric signals; Penetration states identification; Deep learning

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## 1. Introduction

Laser welding technology has been widely applied in industrial manufacturing due to its high energy density, high production efficiency, ease of automation, and minimal thermal deformation [1], [2]. But the complicated laser-metal interaction can negatively impact

4. Results and discussion

5. Conclusions

6. Future works

CRediT authorship contribution statement

Declaration of competing interest

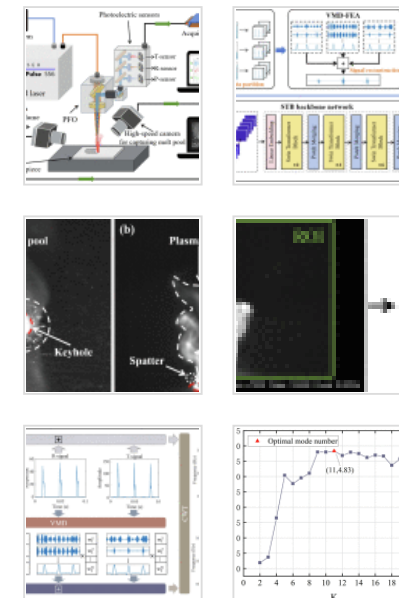
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weld quality and stability, which may lead to several defects, including shallow or deep penetration, high-levels of porosity, and metal spatters [3], [4], [5]. To ensure optimal product quality, it is crucial to achieve reliable online monitoring and process control for the laser welding [6], [7], [8]. Among the various physical phenomena resulting from laser-metal interaction, optical radiation is a prominent physical phenomenon during laser deep penetration welding. Its multi-waveband signals are closely associated with the welding process and quality, which is vital for online monitoring and identification of welding defects [9]. Experimental results reveal that optical radiation signals in the laser welding process are mainly composed of ultraviolet and visible light wavebands (300–700nm), laser reflection wavebands (fiber laser 1064–1070nm), and infrared radiation wavebands (1100–1600nm) [10]. Capturing the optical signal characteristics of the laser welding process is essential, as it contains a substantial amount of information regarding the welding quality. By combining photodiode sensors with different optical filter modules, plasma radiation, thermal emission, and laser reflection can be efficiently detected [11]. Therefore, after determining the target spectral bands, it is even more important to discern effective information within the multi-spectral optical signals.

In recent years, research on optical signal monitoring of laser welding processes has made notable progress [12]. Bono et al. claimed that photodiode signals obtained during the laser welding process can be used to infer process anomalies related to incorrect laser power settings, poor joint preparation, and joint gap [13]. In their study, they deliberately altered the laser power, joint preparation, cleanliness, and fit-up, and found that the photodiode signals, ranging from 600 to 850nm, exhibited significant fluctuations at the locations of anomalies. Although this research revealed a significant correlation between different types of process anomalies and photodiode signal behavior, it remains critically important to qualitatively classify signals based on process anomalies to develop more robust monitoring systems. Chianese et al. have utilized photodiode signals to detect variations in component gaps and changes in weld penetration depth [14]. In their study, based on the proposed time-domain features, such as energy intensity and scatter level, they found a strong correlation between the plasma plume light signals in the 300–

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700nm range and the temperature signals in the 1200–2000nm range with laser power and component gaps. The authors further concluded that characterizing the photodiode signals can effectively determine changes in weld quality. On the one hand, the signal can be analyzed in the time domain, while on the other hand, frequency analysis can be applied. Schmidt et al. identified distinct frequency regions for the oscillations of the melt pool (300–500Hz) and keyhole (2000–2500Hz) [15]. Zhang et al. pointed out that keyhole area and centroid high plasma plume vary differently in the time domain but similarly in the frequency domain [16]. In different penetration states, the low-frequency component (0–4000Hz) is the dominant data, and the frequency amplitude effectively distinguishes between partial and full penetration states.

In summary, numerous studies have described the characteristics of optical radiation signals from both time-domain and frequency-domain perspectives to reveal their correlation with welding conditions. However, due to the limitations of the scene and the diversity of feature recognition algorithms, extracting features from sensor data based on empirical knowledge faces challenges in terms of universality and efficiency. Consequently, many studies have shifted focus to analysis methods driven by the signals themselves, such as Empirical Mode Decomposition (EMD) and Variational Mode Decomposition (VMD) [17], [18], [19]. These adaptive signal analysis techniques have been proven to effectively process sensor signals during the welding process. For instance, Fan et al. employed EMD to decompose the time series of visual features in the laser-metal inert gas hybrid welding process, resulting in multiple intrinsic mode functions [20]. Unlike other studies, these mode components were directly used as inputs for a Support Vector Machine (SVM) to identify welding defects, including sound welding, root humping, incomplete penetration, and burn through. The results show that the introducing the EMD algorithm enables better classification of defect types. Chen et al. applied VMD to the vibration signal analysis of ultrasonic metal welding [21]. Their results showed that the effective vibration signals could be optimally separated from the interference using VMD. Based on the effective vibration signals, they achieved excellent prediction of the contact resistance and tensile strength of the wire terminals. Similarly,

to handle acoustic emission signals during laser welding, Luo et al. proposed an adaptive time–frequency technique based on VMD to characterize the acoustic energy distribution between sub-signals with low-frequency and high-frequency components under different welding penetrations [22]. Their study showed that the VMD can not only effectively remove the noise of acoustic emission signals but also significantly improve the accuracy of the weld seam depth classification model. Chen et al. applied the VMD to the spectral signal analysis of the Gas Tungsten Arc Welding (GTAW) process, coupled with correlation coefficients calculation, which significantly reduced redundant information in the spectral data [23]. They further noted that the VMD resolved issues of mode number uncertainty and modal aliasing caused by the EMD. Compared to EMD, VMD has a narrower bandwidth and stronger physical significance, making it more suitable for real-time analysis of non-stationary optical radiation signals. Considering these findings, VMD appears to be of considerable importance in providing critical information for monitoring the laser welding process. However, due to the limited number of studies, a deep understanding of the relationship between modal components from optical radiation signals and the penetration depth achieved in pulsed laser welding remains challenging. Moreover, VMD has been extensively explored in areas like mechanical fault diagnosis, environmental monitoring, and biomedical signal processing, indicating its significant potential for further development [24], [25], [26].

Based on the aforementioned studies, it can be summarized that many research efforts to date have attempted to explain the significant correlation between welding quality and various sensor signals. Monitoring, as a crucial component of automated welding closed-loop control, is vital in ensuring accuracy. However, the precision of monitoring depends on how statistical parameter are represented, and shallow structures cannot capture high-level representations of signals from large datasets. In recent years, due to their powerful identification and classification capabilities, deep learning methods have gained attention and been successfully applied to monitoring frameworks [27], [28]. Zhang et al. proposed a framework for in-situ judgement based on photoelectric sensors and deep learning [29]. The reflected laser radiation and visible light radiation were captured, and

the signals were analyzed and processed using the wavelet packet decomposition for obtaining time–frequency spectrum diagrams. Finally, a Convolutional Neural Network (CNN), a type of deep learning model, was used to identify the spectrograms spliced from the two optical signals, achieving online judgement of welding quality. Cao et al. adopted a photoelectric sensor and an acoustic sensor to collect welding signals and proposed a cross-attention fusion CNN to capture signal characteristics for welding quality classification [30]. Liu et al. used a visual monitoring system to acquire images of the melt pool under different penetration states from multiple viewpoints and implemented a CNN-based model to predict the weld depth [31]. To sum up, the significance and importance of CNN-based models in the application of laser welding process monitoring are profound.

While CNN models remain dominant, it is noteworthy that a new neural architecture called the vision transformer (ViT), has demonstrated superior performance over CNN in multiple tasks within the field of computer vision [32]. It not only surpass many popular CNN models in terms of computational efficiency and accuracy but also holds considerable potential for future development [33]. Concurrently, some studies have begun to apply the ViT architecture to welding monitoring. For instance, based on the ViT model, Liu et al. proposed an improved and optimized framework for welding defect detection and identification [34]. This framework, which takes weld images as input, is capable of identifying common welding defects, including burr, concave, and porous. Their research indicates that the ViT architecture significantly enhances the model's ability to extract global features from weld images, resulting in a notable performance improvement. Wang et al. have attempted to apply various architectures of ViT models to the recognition of weld penetration in GTAW, confirming the feasibility of their application [35]. They further pointed out that compared to other conventional recognition methods, ViT models, due to their self-attention mechanism, possess higher accuracy and stronger generalization performance. Compared to ViT, the shifted-windows-based (Swin) transformer, a newer transformer architecture, has lower computational complexity [36]. At the same time, the Swin transformer introduces a

hierarchical construction method commonly used in CNNs, allowing it to serve as a general-purpose backbone network. However, despite the growing advantages of transformer architectures, they have not yet been widely applied to laser welding monitoring.

In this work, our aim is to introduce a novel approach for online monitoring of laser welding. This method is expected to enhance monitoring efficiency and accuracy. It will also provide a solid basis for adaptive control of weld quality. To address these challenges, our contributions are summarized as follows:

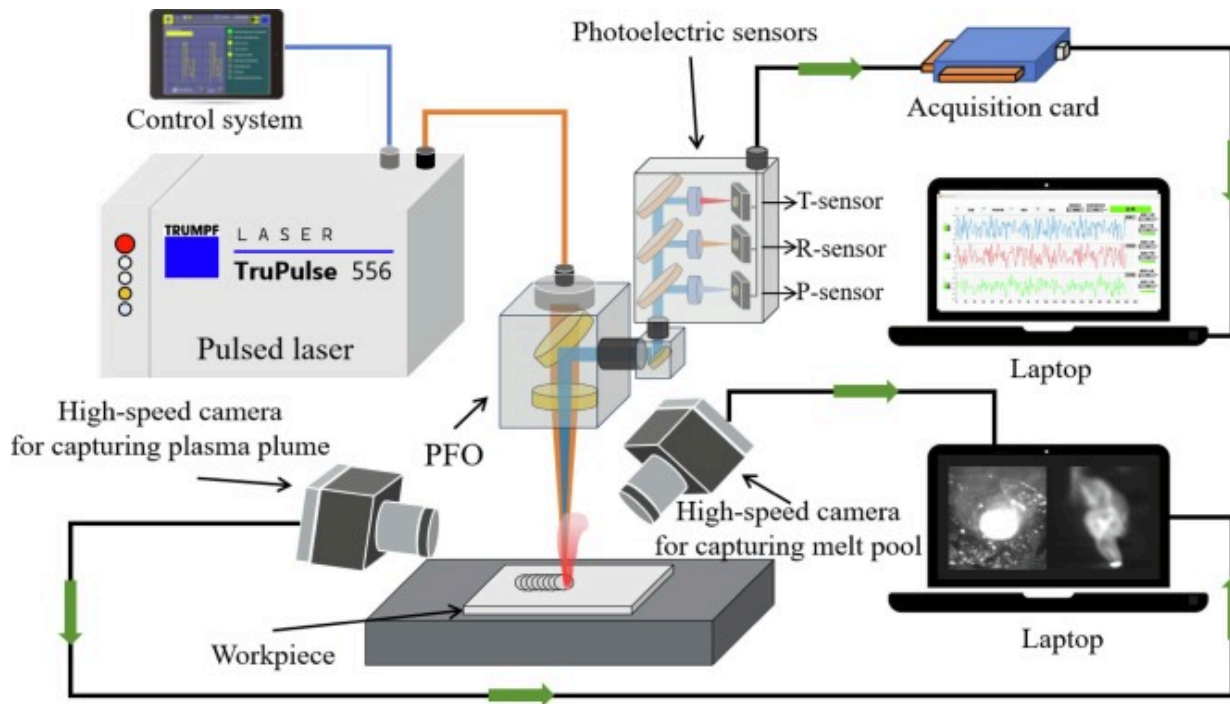
- (1) By investigating the transient properties of plasma and their correlation with optical radiation signals, the photoelectric sensing technique is able to more accurately characterize keyhole dynamics and weld penetration depth. It will also provide crucial guidance for understanding the complex interaction mechanisms between pulsed laser radiation and aluminum alloys, which can be extended to other laser-based manufacturing scenarios.
- (2) Compared to traditional time-domain or frequency-domain feature extraction, this paper introduces an adaptive analysis technique called Variational Mode Decomposition and Frequency-domain Energy Analysis (VMD-FEA), which can accurately obtain modal components significantly related to penetration depth, thereby better characterizing weld penetration depth. Moreover, this method can be applied to other sensor signal analyses in the welding field.
- (3) Based on the Swin transformer architecture, a novel penetration state identification model, STN, is proposed. This model delves into the spatio-temporal features of optical signals from two-dimensional signal time–frequency diagrams. It enhances the performance of penetration depth prediction based on photodiode sensing, providing a reliable monitoring technology for accelerating the intelligence and automation of industrial manufacturing and improving the quality of laser welding.

This paper is organized as follows: [Section 2](#) describes the structure of the experimental platform, while [Section 3](#) details the data processing process and model architecture of this study. In [Section 4](#), the results of the data analysis and prediction evaluation of the model are shown. Finally, [Section 5](#) summarizes the conclusions and future plans of this study.

## 2. Experimental setup

As shown in [Fig. 1](#), the experimental monitoring platform consists of a pulsed laser system, a high-speed photography system, and a photoelectric signal collection system. The pulsed laser system is composed of a pulsed laser, programmable focusing optics (PFO), and a control system. The pulsed laser (TRUPULSE 556) from TRUMPF, a high-powered laser system, has a power of 530W, a wavelength 1064nm, a max peak pulse power of 20kW, pulse width from 0.3 to 50ms, and pulse energy ranging from 60 to 100J. PFO enables the laser beam to be positioned at any specified location within the processing field or space or along any seam geometry. The control system adjust the motion parameters during the welding process. High-speed photography is used to collect plasma plume and melt pool information. The photoelectric sensing system consists of three sensors: the T-sensor, R-sensor, and P-sensor. The T-sensor comprises a photodetector (PDA20CS2, THORLABS) with a detection range of 800–1700nm and a bandpass filter (1100–2100nm) to detect near-infrared radiation. The R-sensor includes a photodetector (PDA100A2, THORLABS) with a detection range of 320–1100nm and a narrowband filter ( $1064\pm 2$ nm) to detect the laser reflected by the material. The P-sensor consists of the PDA100A2 photodetector and a bandpass filter (300–700nm) for detecting visible light radiation. All three sensors are connected to a data acquisition card, with the sampling frequency set to 40kHz. The high-speed camera used is the Phantom VEO 710L, equipped with a macro lens (AF Micro-Nikkor ED 200mm f/4D IF, Nikon). The lens aperture is set to F16, with a resolution of  $512\times 512$ , a sampling rate of 15,000 fps, an exposure time of 20 $\mu$ s, and an exposure index (EI) set to 6400. When capturing images of

the molten pool, an 810nm long-pass filter and an active light source (CAVILUX HF, CAVITAR) are attached to the lens to reduce interference from high-intensity radiation and optimize imaging quality.



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Fig. 1. Plasma plume and spectral monitoring experimental platform for laser welding.

During the laser welding process, the control system sends a signal to the TRUPULSE 556 pulsed laser. The laser receives the command to generate a laser beam, and after a delay of 10s, the laser beam is transmitted through a fiber optic via the PFO to the surface of the workpiece. During this delay, the high-speed camera and photoelectric sensors are activated. Plasma plume information and photoelectric signals are stored in laptop computers. The monitoring platform enables real-time acquisition of plasma plume and photoelectric signals simultaneously. The experimental materials used was 1060

aluminium alloy with a thickness of 1 mm. Welding experiments were performed using laser power as a variable to obtain welds with different penetration depths. [Table. 1](#) shows the detailed welding parameters.

Table 1. Detailed welding parameters.

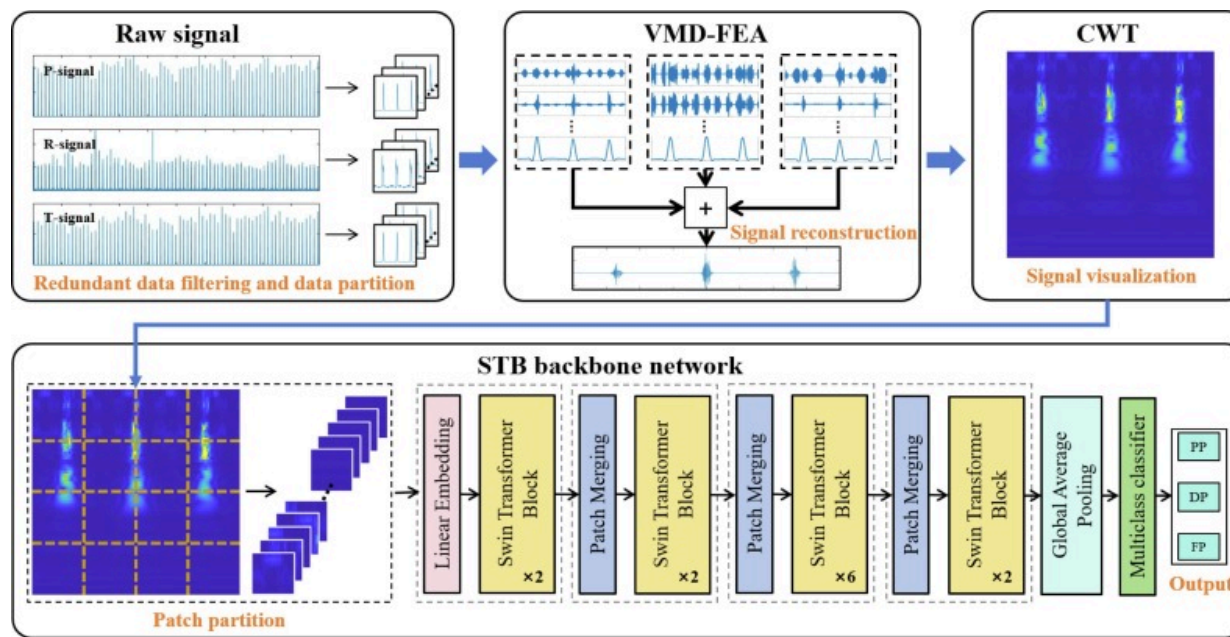
| Case | Changed Parameter (Laser Power, kW) | Unchanged Parameters           |
|------|-------------------------------------|--------------------------------|
| 1    | 3                                   | <b>Pulse duration:</b> 4ms     |
| 2    | 3.4                                 | <b>Pulse frequency:</b> 30Hz   |
| 3    | 3.7                                 | <b>Welding speed:</b> 7.25mm/s |
| 4    | 3.8                                 | <b>Defocus distance:</b> 0mm   |
| 5    | 4                                   |                                |
| 6    | 4.5                                 |                                |

### 3. Methods

#### 3.1. Overall framework

The overall structure of the monitoring strategy, which is based on signal decomposition-reconstruction and the Swin transformer model, is shown in [Fig. 2](#). First, the photoelectric signals of three different wavelengths during the welding process are collected in real time through the photoelectric signal acquisition system. Furthermore, photoelectric sensors collect a large amount of redundant data during the intervals of pulsed laser emission. Therefore, data segmentation is necessary to alleviate storage pressure. The data is then segmented into fixed-length windows, each containing 4,000 data points. This segmentation ensures that each window includes valid signals from three laser pulses, while avoiding excessive computational load due to overly long data sequences.

Second, to establish the correlation between the signal and the penetration state, the photoelectric signal is decomposed and reconstructed through the variational modal decomposition and frequency-domain energy analysis (VMD-FEA). The reconstructed signals are then converted into 2D images using continuous wavelet transform (CWT). Third, patch partition is applied to the image, which is then fed into a backbone network composed of Swin transformer blocks (STBs) to extract features. Finally, a fully connected layer serves as a multi-class classifier to identify weld penetration states from the extracted feature vectors.



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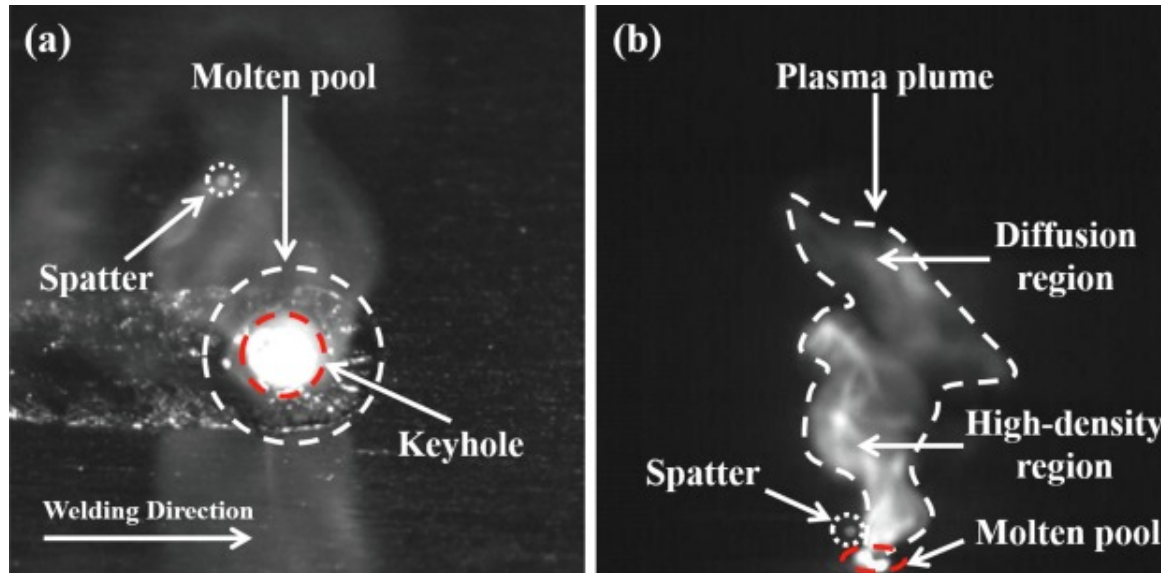
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Fig. 2. The overall framework of the proposed monitoring strategy.

### 3.2. High-speed photographic image processing methods

During the welding process, the transient changes in the molten pool and plasma plume directly impact the weld quality. To closely observe the behavior of the molten pool and

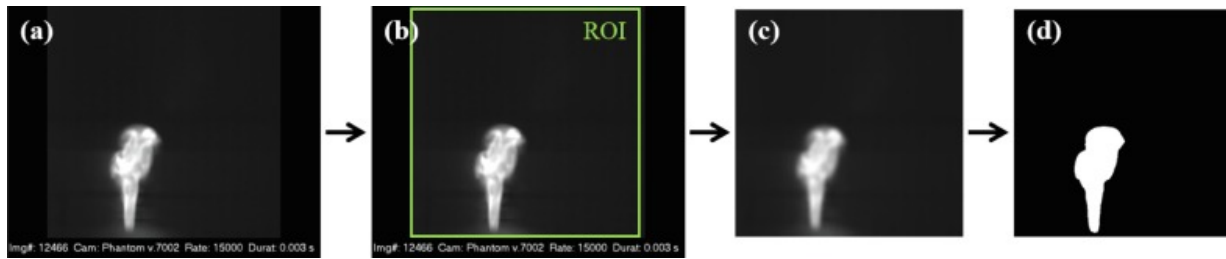
plasma plume formation, high-speed photography is employed to capture their dynamic process. Typical images obtained are shown in Fig. 3. Due to the high energy density of the laser, small spatter is generated near the keyhole. A plasma plume is observed above the molten pool and consists of two parts: a high-density region and a diffusion region. Outward diffusion of the plasma plume results in a blurred boundary. Fig. 4 shows the entire contour extraction process for a typical plasma plume image. Fig. 4 illustrates the high-speed imaging conducted to analyze the dynamic behaviors of the plasma plume. The area of the plasma plume was considered a characteristic parameter for evaluation. Fig. 4(a) presents the original image captured by the high-speed camera, while Fig. 4(b) displays the extracted Region of Interest (ROI), with dimension of  $512 \times 512$ . Fig. 4(c) demonstrates the application of Gaussian filtering to the ROI to smooth the plasma edges. Fig. 4(d) depicts the resultant image after applying a 0.28 threshold binarization, yielding a well-defined plasma image. All plasma plume images were set with 0.28 as the binarization threshold, and the contours were extracted based on this threshold to calculate the area.



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Fig. 3. Examples of high-speed photographic images: (a) molten pool; (b) plasma plume.



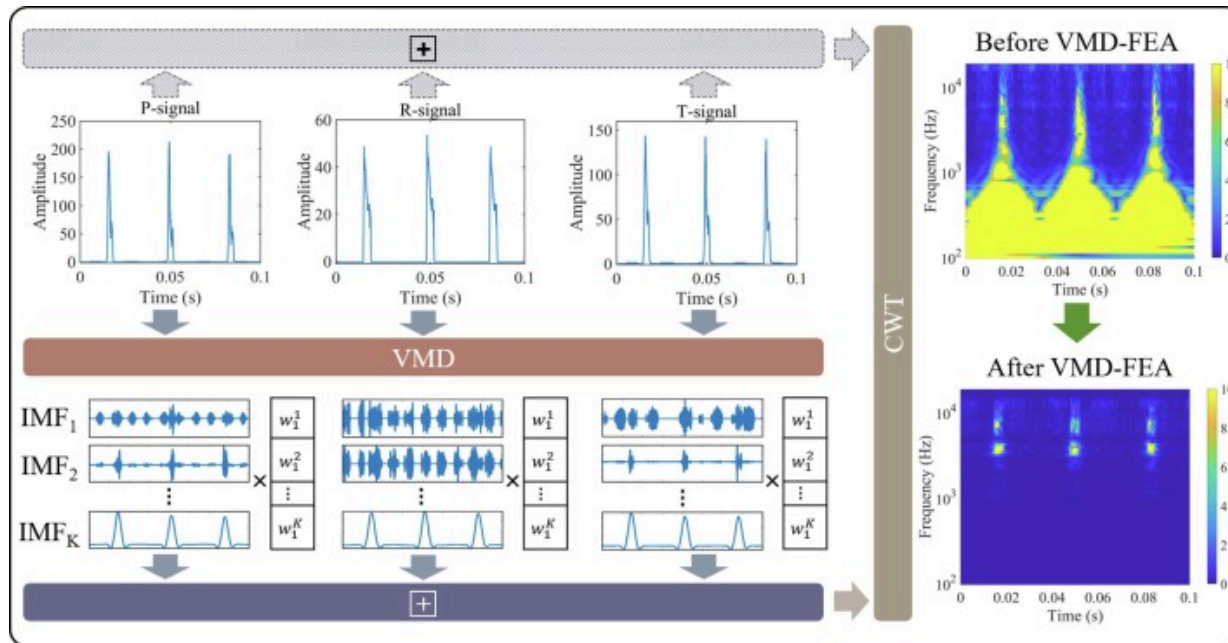
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Fig. 4. Analysis process of a plasma image: (a) original image, (b) region of interest, (c) Gaussian filtered image, (d) plasma segmentation image.

### 3.3. Photoelectric signal processing method

The photoelectric signal encompasses the abundant optical radiation from the welding process. Because of the high sensitivity and sampling frequency of photoelectric sensors, it is essential to reduce noise and identify key features. Fig. 5 illustrates the overall processing of photoelectric signals used in this study, which primarily involves signal decomposition, reconstruction, and visualization.



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Fig. 5. Photoelectric signal processing processes.

### 3.3.1. Signal decomposition

In the welding process, the T-sensor, R-sensor, and P-sensor are synchronized to collect photoelectric signals across three distinct wavelength ranges. To develop a reliable model for identifying penetration states, it is crucial to uncover data features that correlate with the weld prediction depth. VMD decomposes the signal into multiple Intrinsic Modal Functions (IMFs), which helps separate different frequency components and enhances the signal-to-noise ratio. The VMD algorithm helps pinpoint the most significant modal components within the multi-level frequency spectrum of the signal.

The essence of VMD lies in formulating and solving a variational problem to minimize the cumulative bandwidths of the IMFs. This must be done while ensuring that the sum of all

IMFs reconstructs the original signal. Suppose the original signal is  $f(t)$ , which is decomposed into  $K$  IMFs. These IMFs are denoted as  $u_k(t)$ , with the center frequency of each IMF being  $\omega_k$ . Then the objective function of VMD is:

$$J = \alpha \sum_{k=1}^K \|\partial_t \left[ \left( \delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t}\|_2^2 + \|f(t) - \sum_{k=1}^K u_k(t)\|_2^2 \quad (1)$$

where  $\alpha$  is a regularization parameter,  $\delta(t)$  is the Dirac delta function, and  $*$  denotes convolution. The symbol  $\partial_t$  denotes differentiation with respect to time  $t$ , and  $\|\bullet\|_2$  denotes the  $L_2$  norm, which is commonly used to measure the magnitude of a function in signal processing. The constraint of VMD is:

$$\sum_{k=1}^K u_k(t) = f(t) \quad (2)$$

To solve this variational problem, the augmented generalized Lagrange multiplier method (ADMM), a powerful algorithm for solving constrained optimization problems, is employed by introducing a Lagrange multiplier  $\lambda(t)$  and alternately updating  $u_k(t)$ ,  $\omega_k$  and  $\lambda(t)$  until convergence. The specific update formula is:

$$\begin{cases} u_k^{n+1}(t) = \operatorname{argmin}_{u_k} J \\ \omega_k^{n+1} = \operatorname{argmin}_{\omega_k} J \\ \lambda^{n+1}(t) = \lambda^n(t) + \rho \left( f(t) - \sum_{k=1}^K u_k^{n+1}(t) \right) \end{cases} \quad (3)$$

where  $\rho$  is a positive penalty parameter and  $n$  denotes the number of iterations.

To effectively decompose the photoelectric signal with the VMD algorithm, it is essential to first determine the optimal mode number,  $K$ . This study is inspired by Li et al.'s method, optimizes the VMD algorithm using the envelope kurtosis maximization principle, and identifies the optimal mode number  $K$  from the range of 2 to 20 [24]. Firstly, the envelope of each IMF is calculated using the subsequent formula:

$$x t_K^i = \left| \frac{1}{\pi t} \bullet x_K^i(t) \right|, i = 1, 2, \dots, K \quad (4)$$

where  $x_K^i(t)$  denotes the  $i$ -th IMF produced by VMD with a mode number of  $K$ , and  $xt_K^i$  is the absolute value of the Hilbert transform of the  $i$ -th IMF.

Additionally, the envelope kurtosis for the  $i$ -th IMF, when  $K$  is the mode number, is computed as follows:

$$ek(i) = \frac{E(xt_K^i - \mu(xt_K^i))^4}{\sigma(xt_K^i)^4}, i = 1, 2, \dots, K \quad (5)$$

where  $\sigma(xt_K^i)$  represents the standard deviation of  $xt_K^i$ , and  $E(xt_K^i - \mu(xt_K^i))^4$  denotes the fourth-order central moment of  $xt_K^i$ .

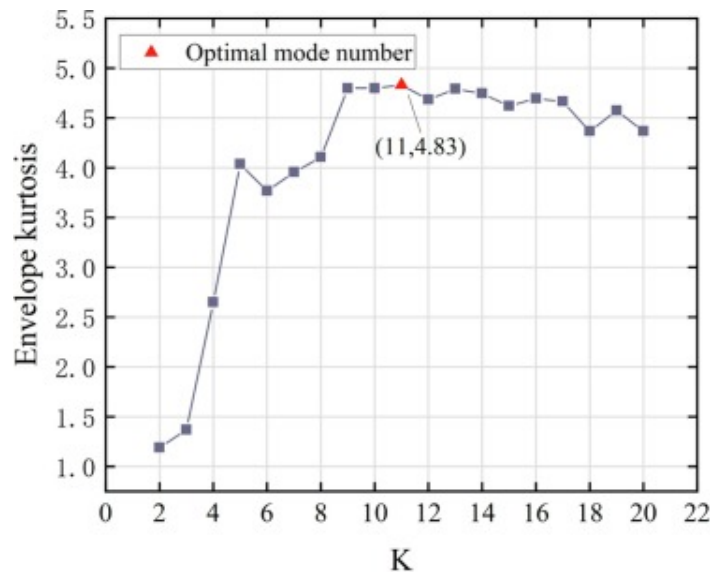
For a mode number of  $K$ , calculate the envelope kurtosis for all IMFs using to Eq (5), and identify the maximum value  $ek_K^{max}$ , as follows:

$$ek_K^{max} = \max(ek_1, ek_2, \dots, ek_K) \quad (6)$$

Given a search range for  $K$  from 2 to 20 with a step size of 1, this results in 19 local maximum values over the entire search interval. The global maximum,  $ek_{global}^{max}$ , is then determined based on these 19 local maximum values:

$$ek_{global}^{max} = \max(ek_2^{max}, ek_3^{max}, \dots, ek_{20}^{max}) \quad (7)$$

Finally, the optimal mode number  $K$  is determined based on the global maximum envelope kurtosis value  $ek_{global}^{max}$ . The relationship between the mode number  $K$  and the maximum value of envelope kurtosis within the search domain is illustrated in Fig. 6. When  $K = 11$ , the maximum value of envelope kurtosis is calculated to be 4.83, leading to the selection of the optimum mode number,  $K = 11$ .



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Fig. 6. The relationship between K and the maximum of envelope kurtosis.

### 3.3.2. Signal reconstruction

To assess the IMF components derived from the original signal, the frequency domain energy (FE) of each component is selected as the evaluation metric. The FE is typically defined as an indicator of signal strength and is calculated as follows:

$$X_i^c(k) = \sum_{n=0}^{N-1} IMF_i^c(n) e^{-j \frac{2\pi}{N} kn}, k = 0, 1, \dots, N-1 \quad (8)$$

$$FE_i^c = \sum X_i^c(k)^2 \quad (9)$$

where  $IMF_i^c$  denotes the  $i$ -th IMF component of the original signal from channel  $c$ ,  $X_i^c(k)$  represents the frequency-domain signal obtained subsequent to Eq. (8),  $N$  is the signal length,  $k$  is the frequency index,  $j$  represents the imaginary unit, and  $FE_i^c$  is the associated frequency energy. To assess the correlation between the reconstructed signal

and the prediction depth, the Pearson correlation coefficient  $r_{FE_i^c, D}$  between the FE and the penetration depth  $D$  is used as an evaluation metric, calculated as follows:

$$r_{FE_i^c, D} = \frac{\sum_{n=1}^N (FE_{i_n}^c - \bar{FE}_i^c)(D_n - \bar{D})}{\sqrt{\sum_{n=1}^N (FE_{i_n}^c - \bar{FE}_i^c)^2 \sum_{n=1}^N (D_n - \bar{D})^2}} \quad (10)$$

The weight assigned to each IMF in the signal reconstruction process is determined by the value of  $r$  according to the following decision criteria:

$$\begin{cases} If |r_{FE_i^c, D}| \geq 0.9, w_i^c = 1 \\ If |r_{FE_i^c, D}| < 0.9, w_i^c = 0 \end{cases} \quad (11)$$

Based on  $w_i^c$ , the signal reconstruction is performed, and the calculation formula is as follows:

$$Y_i^c = \sum_{k=1}^K IMF_k^c * w_i^c \quad (12)$$

where  $Y_i^c$  is the reconstructed signal.

### 3.3.3. Visual processing of signals

Once the fusion of the three channel signals is complete, the signals are transformed from the time domain to the time–frequency domain using the continuous wavelet transform (CWT) for visualization. The CWT formula is as follows:

$$CWT(s, t) = \frac{1}{\sqrt{|s|}} \int_{-\infty}^{+\infty} x(\tau) \psi^* \left( \frac{\tau - t}{s} \right) d\tau \quad (13)$$

where  $x(\tau)$  is the signal,  $\psi(\tau)$  is the wavelet function,  $s$  is the scaling factor,  $t$  is the translation factor, and  $\psi^*$  is the complex conjugate. The result of the CWT is a complex matrix that represents the wavelet coefficients of the signal at different scales (the inverse of the frequency) and times points. This matrix can be used to create a time–frequency representation of the signal, illustrating the frequency variations over time. To construct the deep learning model's dataset, the time–frequency graphs are divided into

three categories based on the corresponding penetration depths, with each category containing 1240 images.

### 3.4. Penetration identification model based on Swin transformer

To detect penetration states, a Swin transformer network (STN) utilizing Swin transformer blocks (STBs) as its backbone is designed to extract the most discriminative features of different penetration states from time-frequency diagrams. Although CNN and its variants remain the primary backbone architectures for computer vision applications, transformers offer a promising alternative for vision tasks. Transformers have a larger respective field, allowing the network to consider a broader range of contexts, which is crucial for understanding complex patterns and relationships. In CNNs, a hierarchy of local features is used to represent an image, but this limits the respective fields, requiring the CNN to aggregate global information by stacking more convolution layers [37]. In contrast, transformers utilize the self-attention mechanism to capture spatial patterns and non-local dependencies, enabling the network to obtain rich global information with fewer transformer blocks. Furthermore, a larger receptive field helps average out local variations and noise, facilitating the network's ability to capture more robust features from complex multi-photoelectric signals.

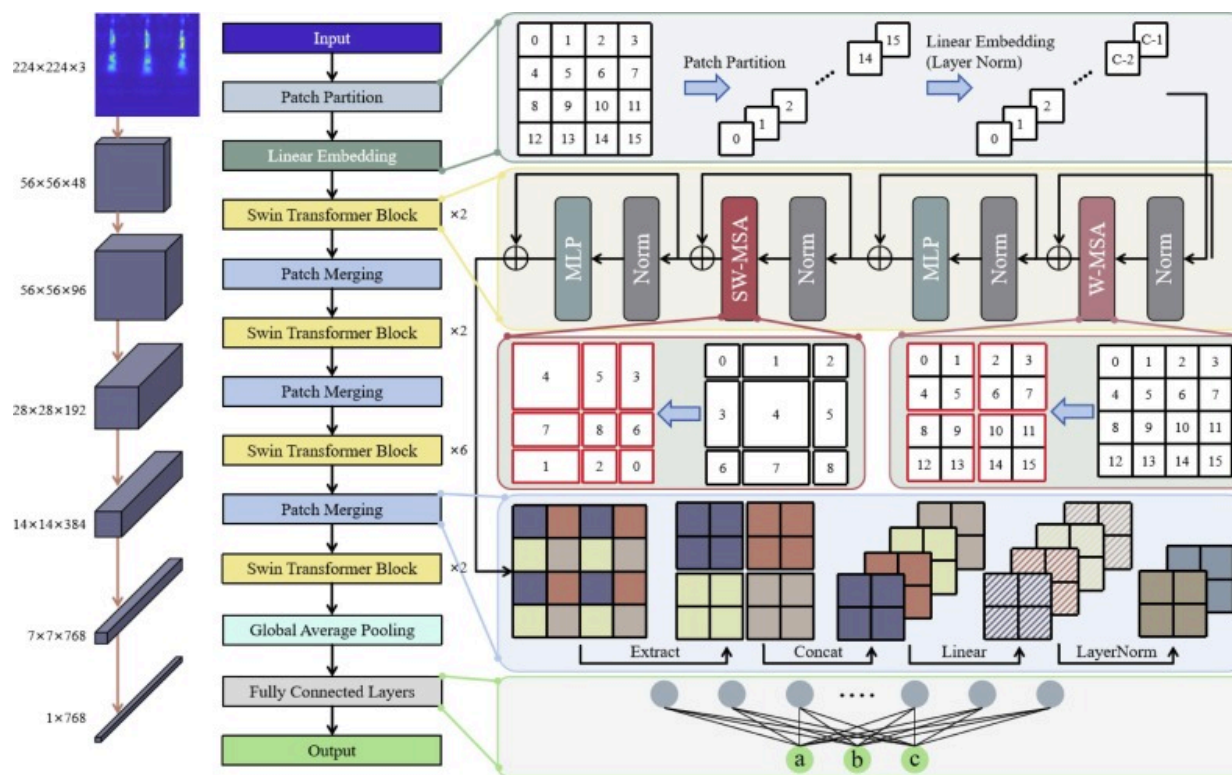
[Fig. 6](#) illustrates the architecture and calculation principle of the STN. The model accepts an RGB image input of size  $224 \times 224 \times 3$  pixels. The image is processed by the backbone network, producing an output tensor of dimensions  $7 \times 7 \times 768$ . Then, a global pooling layer compresses this tensor into a  $1 \times 768$  vector, which is passed through a fully connected layer for classification. The model's output indicates one of three possible penetration states: partial penetration (PP), deep penetration (DP), or full penetration (FP). The three elements of the output vector  $[a, b, c]$  from the fully connected layer represent the confidence scores for each of the three states. The category with the highest confidence score is selected as the final prediction.

The Swin transformer segments the input image into several 4x4 patches following the patch partition layer, treating each patch as an 'image block'. A linear embedding layer then converts these patches into C-dimensional vectors, where  $C$  is determined by the hyperparameter. These vectors are then processed by transformer blocks. The Swin transformer consists of a series of transformer blocks, each containing multiple transformer layers. Each layer consists of two sub-layers: a windows-based multi-head self-attention layer known as W-MSA or SW-MSA (for shifted-window-based multi-head self-attention), and a multi-layer perceptron (MLP). The W-MSA layer divides the feature map into several distinct regions, or windows, within which self-attention is calculated, focusing solely on the information inside that window, as defined by the following equation:

$$\begin{cases} \text{Attention}(Q_i, K_i, V_i) = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \\ \text{head}_i = \text{Attention}(Q W_i^Q, K W_i^K, V W_i^V) \\ \text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \end{cases} \quad (14)$$

where  $Q$  represents the query vector, which encapsulates the information the model seeks to extract.  $K$  stands for the key vector, embodying the input's features or information.  $V$  is the value vector, representing the content or information of the input data. The attention mechanism calculates the scaled dot product between  $V$  and  $K$  to measure their similarity. This calculation results in a weight matrix that indicates the significance of each input element, determined by the Softmax function. This matrix is then applied to  $V$ , producing a weighted sum that constitutes the attention output. The MSA within a window is achieved by integrating the attention outputs from individual subspaces. To enhance inter-window information exchange, a shifting technique is incorporated into the SW-MSA algorithm, allowing overlap between adjacent windows. The MLP consists of two densely connected layers, separated by a Dropout layer to prevent overfitting. Each STB consists of two sub-blocks: one incorporating W-MSA and the other incorporating SW-MSA.

Each transformer module ends with a patch merging layer (PML), which reduces the resolution of the feature map and increases the number of channels. As shown in Fig. 7, the color of the patch indicates its location. The feature map is derived based on the patch's position and is then concatenated along the depth axis. After this, normalization is applied to the concatenated feature map, followed by a linear transformation that reduces the dimensionality along the depth axis. PML progressively broadens the receptive field to gather more global information. Following all the transformer modules, a global average pooling layer compresses the feature map into a vector. Finally, a fully connected layer generates the classification outcomes.



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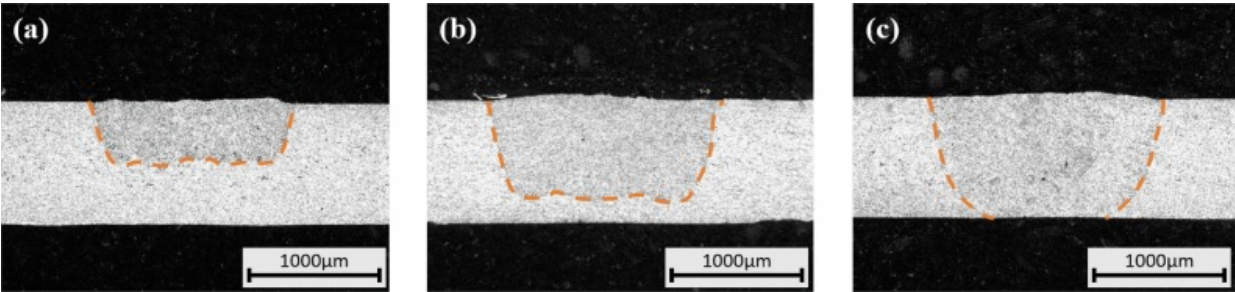
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Fig. 7. The architecture and calculation principle of STN.

4. Results and discussion

4.1. Penetration states

Fig. 8 illustrates the cross-section of seams, demonstrating that different penetration states correspond to varying process parameters. In this study, the penetration state is classified into three types: partial penetration (PP, penetration depth <500μm), deep penetration (DP, 500μm <penetration depth <1000μm), and full penetration (FP, penetration depth >1000μm). Table. 2 provides the experimental parameters and corresponding penetration depth data in detail.



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Fig. 8. Cross-section of weld seams: (a) PP; (b) DP; (c) FP.

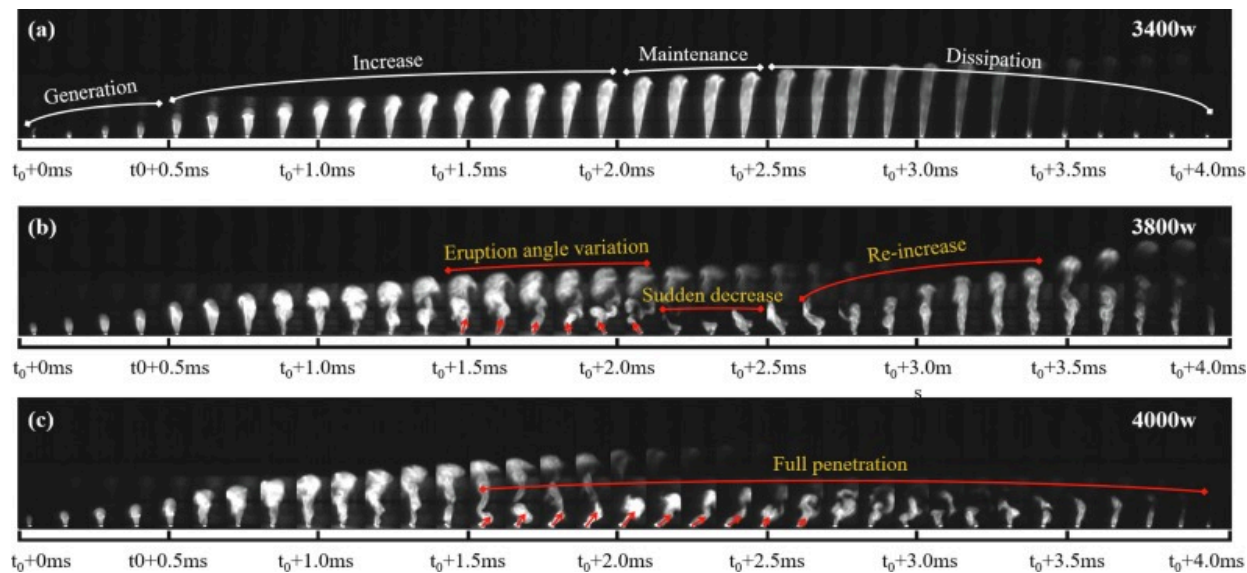
Table 2. Detailed welding parameters and corresponding welding quality.

| Case | Penetration depth (μm) | Penetration states |
|------|------------------------|--------------------|
| 1    | 261.7                  | PP                 |
| 2    | 345.2                  | PP                 |
| 3    | 764.9                  | DP                 |
| 4    | 925.1                  | DP                 |

| Case | Penetration depth ( $\mu\text{m}$ ) | Penetration states |
|------|-------------------------------------|--------------------|
| 5    | >1000                               | FP                 |
| 6    | >1000                               | FP                 |

## 4.2. Plasma plume analysis

**Fig. 9** illustrates the plasma plume eruption, which is segmented into four phases: generation, increase, maintenance, and dissipation. These phase all occur within a 4ms interval. **Fig. 9(a)** depicts the onset of plasma plume generation at time  $t_0$ . During of laser irradiation, the plasma plume area expands. From  $t_0+2.0\text{ms}$  to  $t_0+2.5\text{ms}$ , both the area and brightness of the plasma plume remain at elevated levels. After  $t_0+2.5\text{ms}$ , the plasma plume area progressively diminishes until complete dissipation occurs. Meanwhile, there is no significant change in the geometry or eruption angle of the plasma plume.



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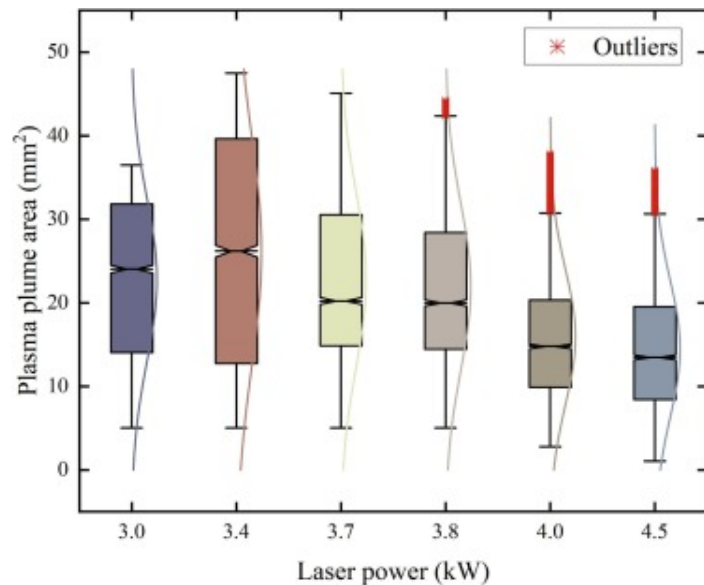
Fig. 9. Plasma plume in the different states: (a) PP; (b) DP; (c) FP.

---

Increasing the peak power of the laser to 3800W causes the seam to transition from the PP state to the DP state. Fig. 9(b) illustrates that the plasma plume covers a larger area and reaches a higher eruption height during the generation stage compared to the increase stage. Between  $t_0+1.5\text{ms}$  and  $t_0+2.0\text{ms}$ , the eruption angle of the plasma plume changes, which modifies its geometry. From  $t_0+2.0\text{ms}$  and  $t_0+2.5\text{ms}$ , the plasma plume area abruptly diminishes. After  $t_0+2.5\text{ms}$ , it starts to increase again, attributed to the dynamic behavior of the keyhole.

Fig. 9(c) demonstrates that the plasma plume area begins to diminish after  $t_0+1.5\text{ms}$  when the peak laser power reaches 4000W. Unlike the DP state, the plasma plume area does not increase again after  $t_0+2.0\text{ms}$ . The high energy input results in full penetration of the weld, leading to small holes penetrating the workpiece. In result, the majority of the plasma plume is expelled through these perforations at the bottom of the workpiece. This outcome reduces both the height and area of the plasma plume that is ejected upwards.

Fig. 10 illustrates that the plasma area progressively enlarges as the laser power increases from 3000W to 3400W. At a power level of 3400W, the plasma achieves its maximum average area of  $26.15\text{mm}^2$ . Concurrently, the seam is in the PP state with a shallow keyhole, allowing for adequate plasma ejection. As the laser power escalates to 3700W, the average plasma area begins to decrease. At this point, the seam transitions to the DP state. When the laser power reaches 4000W, the average plasma area shrinks to  $15.91\text{mm}^2$ . The presence of outliers in the data suggests significant plasma fluctuation. Analysis revealed that there is an 8.7% reduction in average area when transitioning from the PP to the DP state, and a more pronounced decrease of 21.7% from the DP to the FP state.



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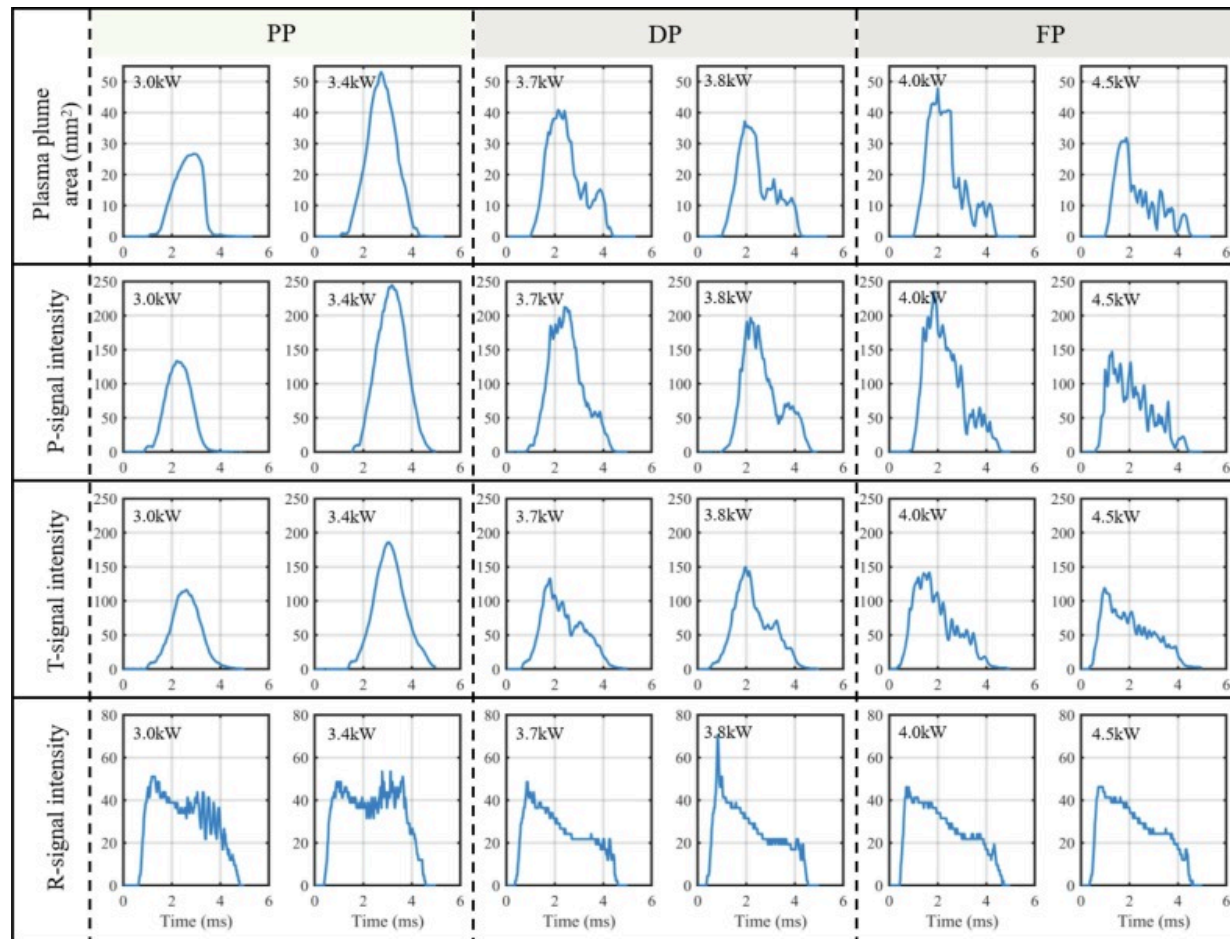
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Fig. 10. Plasma plume characteristics in different penetration states.

### 4.3. Photoelectric signals analysis

To further explore the correlation between high-speed imaging data and photoelectric signals, a comparative analysis was conducted. The plasma plume area during a single pulse was selected and compared with three corresponding photoelectric signals (Fig. 11). In the PP state, the plasma plume area exhibited minimal fluctuations, while in the DP and FP states, significant fluctuations were observed. This aligns well with the plasma dynamics captured in the high-speed imaging data shown in Fig. 9. The analysis reveals a strong correlation between the plasma plume area and the P-signal intensity. This confirms that the P-signal effectively reflects the state of the plasma plume. Additionally, the T-signal shows some correlation with the P-signal, which is consistent with previous findings that the plasma plume contributes to infrared radiation [14]. Notably, both the P-signal and T-signal weaken significantly in the FP state due to the bottom perforation of

the workpiece. Meanwhile, the R-signal remains elevated in the PP state, likely because of lower laser absorption and higher reflection. A stepwise reduction in the R-signal is observed in the DP and FP states as laser absorption increases with penetration depth.

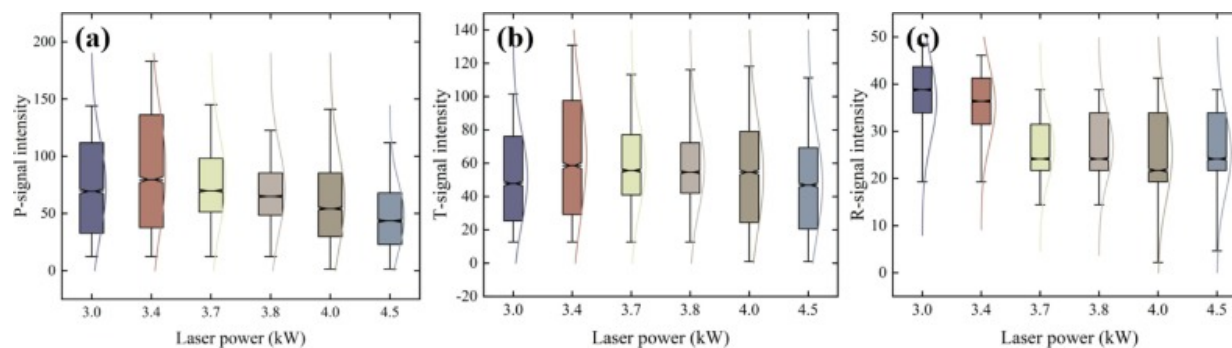


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Fig. 11. Comparative analysis of plasma plume area and photoelectric signal intensity in different penetration states.

Fig. 12(a) demonstrates that the P-signal intensity escalates as the laser power rises within the 3000W to 3400W range. During this period, the welded seam is in the PP state. As the laser power continues to increase, within the 3700W to 3800W range, there is a gradual decline in signal intensity. This coincides with the welded seam transitioning to the DP state. Upon reaching a laser power of 4000W, the signal intensity experiences a continuous reduction. At this point, the welded seam enters the FP state. The transition from the PP to the DP state resulted in a 6.14% increase in average signal intensity. Conversely, the shift from the DP to the FP state led to a 16.1% decrease in average signal intensity. This reduction is attributed to the intense fluctuations of the plasma plume within the keyhole and the influence of bottom perforation (Fig. 9). Only a small portion of the plasma is ejected above the workpiece, resulting in a significant reduction in the optical radiation signal captured by the P-sensor.



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Fig. 12. Photoelectric signals intensity in different penetration states: (a) P-signal; (b) T-signal; (c) R-signal.

With an increase in laser power, more metal is melted due to the laser energy, which correspondingly enhances the contribution to infrared radiation. Fig. 12(b) illustrates a gradual increase in the average intensity of the T-signal. There is a 9.6% increase in the average signal intensity as the weld transitions from the PP state to the DP state. The average signal intensity sees a 10.4% increase when transitioning from the DP state to the

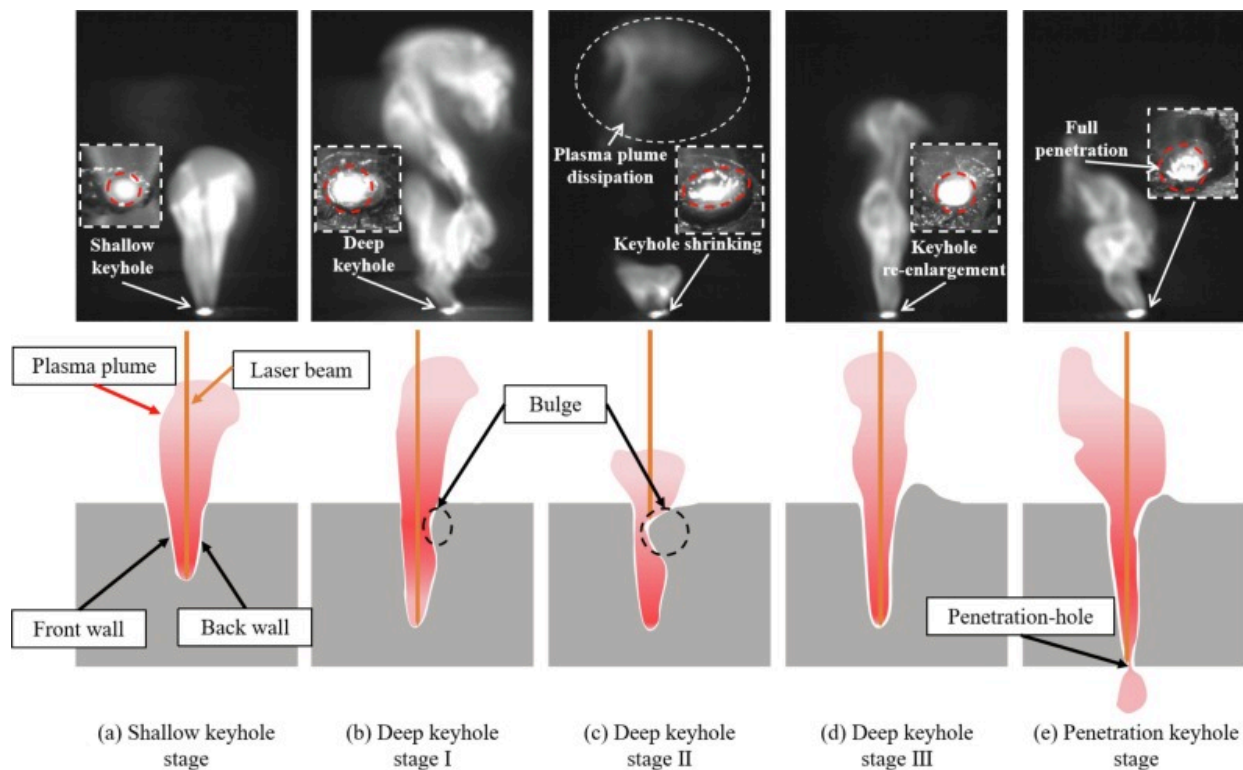
FP state. When the laser power rises to 4000W, the weld seam reaches the FP state, and the average signal intensity drops by 13.6%. As the laser power increases, the P-signal and T-signal exhibited the same trend, which corresponds with the findings of Chianese et al. [14].

Fig. 12(c) demonstrates that the average intensity of the R-signal decreases by 25.3% as the weld transitions from the PP state to the DP state. Research by Olsson et al. [37] indicates that the reflectivity of the weld seam reduces as the curvature of the molten surface increases. As flat weld seam would be the most effective reflector of the laser. In the deep keyhole state, the laser undergoes multiple reflections within the keyhole before exiting and being detected by the sensor, leading to more extensive absorption of laser energy by the workpiece. Additionally, the laser reflection is further reduced due to the perforation at the bottom of the workpiece. The average signal intensity during the FP state is 5.1% higher than that observed in the DP state. The narrow band filter employed by the R-sensor in this study, with a center wavelength of  $1064 \pm 2\text{nm}$  and a full width at half maximum (FWHM) of  $10 \pm 2\text{nm}$ , enables the sensor to monitor a wider range of IR signals. As a result, the signal strength increases slightly.

#### 4.4. The correlation between plasma plume, photoelectric signal and penetration depth

The geometrical features of the plasma plume are intimately connected with the depth of the weld seam, the keyhole depth, and its dynamic behavior. Fig. 13(a) illustrates that, in the PP state, the low peak laser power results in a shallow keyhole depth, a smaller amount of molten metal, and minimal fluctuation in the molten pool. This leads to a smaller plasma plume with a stable shape. In the DP state, the increased laser power is sufficient to create deeper keyholes, and there is a greater likelihood of forming a raised bulge on the keyhole wall, as depicted in Fig. 13(b). With an increase in laser irradiation time, more metal melts, and the bulge grows progressively larger due to the influence of gravity and surface tension. Fig. 13(c) shows how the bulge constricts the keyhole, diminishing its opening and effectively 'locking' the plasma plume within. Nonetheless,

sustained laser irradiation can drive the accumulated plasma plume within the keyhole to be expelled from the protrusion, as demonstrated in Fig. 13(d). In the FP state, depicted in Fig. 13(e), the keyhole perforates the workpiece, allowing the plasma plume to be ejected through the perforations at its base. Consequently, the volume of the plasma plume displays an initial increase followed by a decrease as the power escalates, as evidenced in Fig. 12.



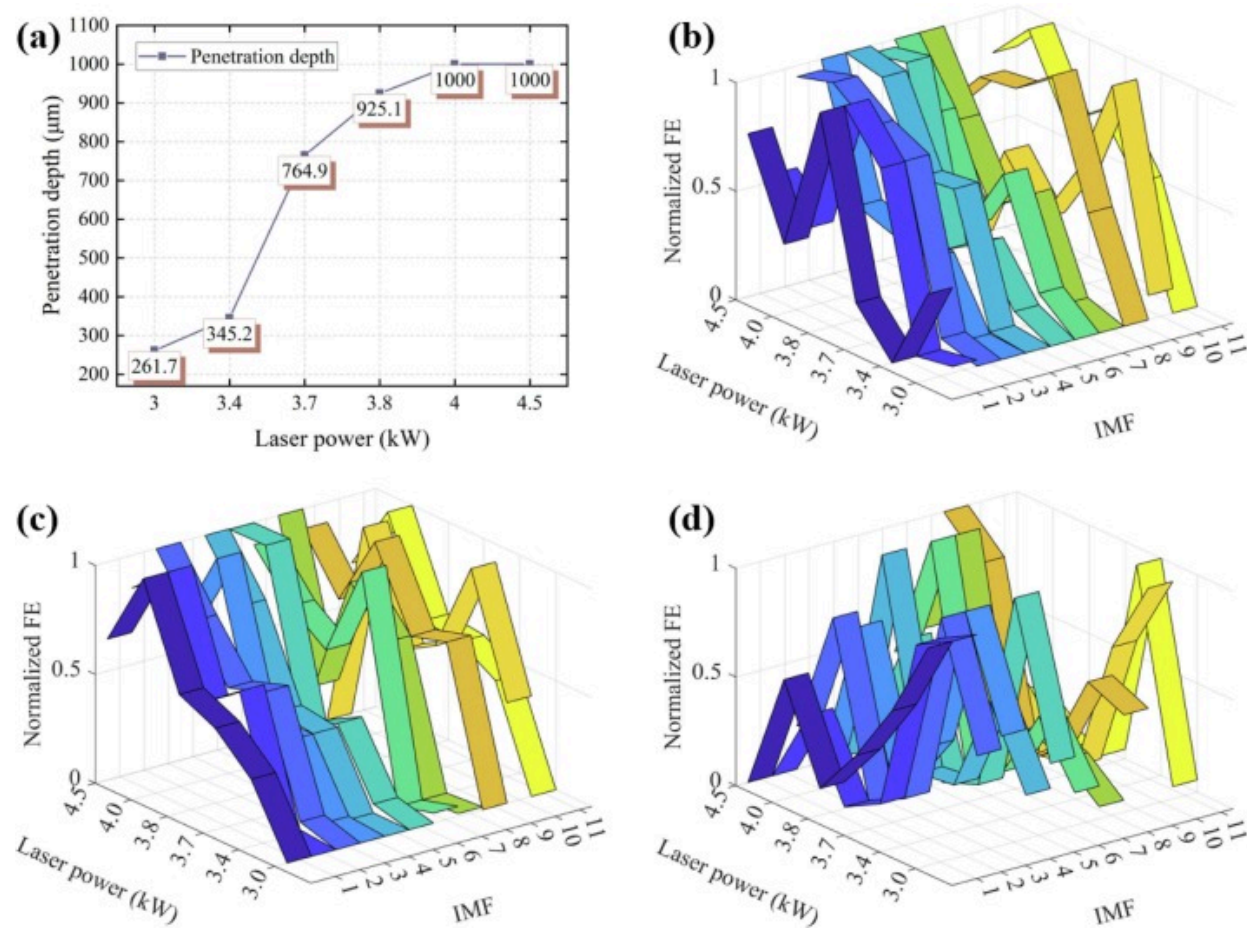
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Fig. 13. Evolution of plasma eruption methods with different penetration states: (a) PP; (b) DP; (c) DP; (d) DP; (e) FP.

Comparing Fig. 10 with Fig. 12(a), one can observe that the variation pattern of the P-signal closely resembles that of the plasma plume volume. Calculations reveal that the

Pearson correlation coefficients between the P-signal and plasma plume volume range from 0.8 to 0.9, indicating that the P-signal effectively reflects the dynamic changes in the plasma plume's physical behavior. To further investigate the characteristics associated with penetration depth in the photoelectric signals, the VMD algorithm was used to decompose the signals for multidimensional analysis. Each IMF component corresponds to a frequency component in the original signal, arranged sequentially from high to low frequency. By analyzing the trend of FE relative to laser power variation for each IMF, it becomes evident that certain components in the P-signal ([Fig. 14\(b\)](#)) and T-signal ([Fig. 14\(c\)](#)) are positively correlated with the penetration depth ([Fig. 14\(a\)](#)). As illustrated in [Fig. 14\(b\)](#), the Pearson correlation coefficient for the IMF3 component derived from the P-signal and related to the penetration depth data reaches a high value of 0.991. This component, associated with a distinct frequency, serves as an effective discriminator between different penetration states, enabling precise differentiation between the critical state of Case4 and the FP state. On the contrary, for some of IMFs in the R-signal, such as IMF1 and IMF2, the FE showed a decreasing trend as the laser power increased ([Fig. 14\(d\)](#)). [Table. 3](#) illustrates that the most components in the P-signal and T-signal exhibit positive correlation with penetration depth, whereas those in the R-signal show negative correlation. This can be attributed to the primary sources of the P-signal and T-signal, which are metal vapor and surface heat radiation from the melt pool, respectively. The capacity for metal evaporation is directly related to penetration depth and weld width. The R-signal indicates the material's absorption of laser light, while high-intensity laser reflections suggest poor energy absorption within the keyhole.



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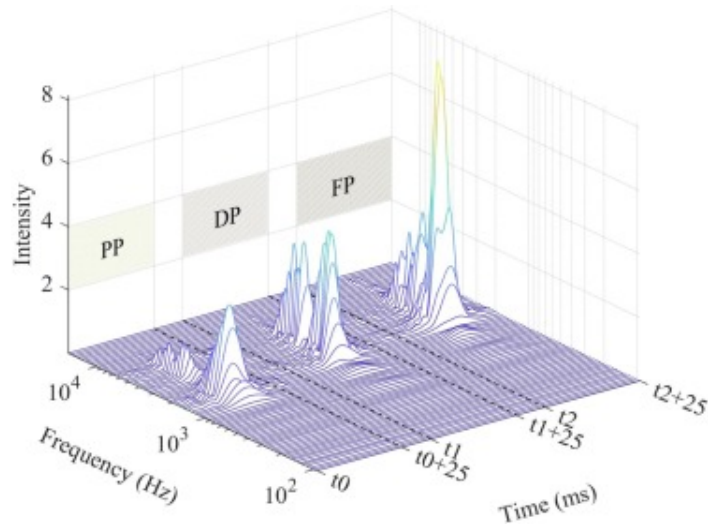
Fig. 14. Weld penetration depth and EF of IMFs with different laser power: (a) penetration depth; (b) P-signal IMFs; (c) T-signal IMFs; (d) R-signal IMFs.

Table 3. Pearson's correlation coefficients between IMFs and penetration depths.

|                 | IMF1    | IMF2    | IMF3    | IMF4    | IMF5   | IMF6    | IMF7   | IMF8   | IMF9   | IMF10   | IMF  |
|-----------------|---------|---------|---------|---------|--------|---------|--------|--------|--------|---------|------|
| <b>P-signal</b> | 0.5602  | 0.7292  | 0.9910  | 0.7941  | 0.9640 | 0.7998  | 0.8374 | 0.8760 | 0.7972 | -0.3570 | 0.85 |
| <b>T-signal</b> | 0.8802  | 0.9376  | 0.9419  | 0.8487  | 0.8726 | 0.8233  | 0.8478 | 0.8018 | 0.7368 | -0.2951 | 0.84 |
| <b>R-signal</b> | -0.8110 | -0.9098 | -0.4425 | -0.5857 | 0.0107 | -0.4949 | 0.2992 | 0.5498 | 0.2056 | -0.8176 | -0.4 |



The intensity, diffusion, and morphological changes of plasma directly impact the stability of the welding molten pool and the uniformity of heat input [38]. By detecting photoelectric signals, especially through the analysis of radiation at specific wavelengths, the dynamic changes of plasma can be monitored in real-time. The amplitude and frequency fluctuations of these signals are closely related to the intensity and stability of the plasma. Experimental results show that the optical radiation signals from plasma can reveal transient phenomena occurring during the welding process. As the laser energy increases, both the plasma area and the intensity of the photoelectric signals follow a similar trend. Additionally, plasma, acting as a bridge between the internal and external parts of the keyhole, indicates unstable keyhole conditions, such as size fluctuations or closure, when it exhibits excessive oscillations. This instability can lead to uneven weld penetration or defect formation [39]. The decomposed signal is reconstructed using Eq. (11) and Eq. (12), and a time–frequency map is generated by applying the CWT transform to the reconstructed signal. Fig. 15 shows that in the PP state, the signal frequency is concentrated between 800 and 2000Hz, with low intensity. In the DP state, the signal frequency is concentrated between 1500 and 6000Hz. In the FP state, the signal frequency is concentrated between 1700 and 6000Hz, with intensity much higher than in the DP state.



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Fig. 15. Time-frequency maps of 3D reconstructed signals in different penetration states.

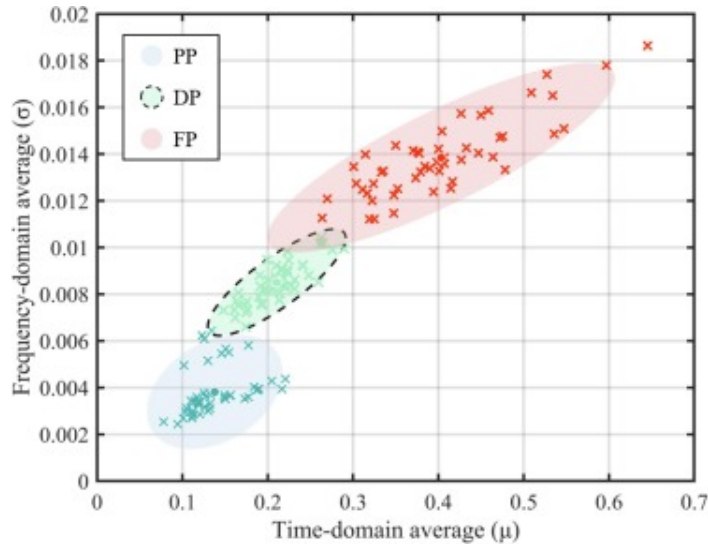
To validate the feasibility of using the ML algorithm for classifying the reconstructed signal  $F_Y$ , two features are extracted:

$$\mu = \frac{\sum_{i=1}^N Y(i)}{N} \quad (15)$$

$$\sigma = \frac{\sum_{i=1}^{N'} F_Y(i)}{N'} \quad (16)$$

$\mu$  represents the time-domain mean of the signal, while  $N$  denotes the signal length.  $\sigma$  represents the frequency-domain mean of the signal.  $F_Y$  is derived from the signal  $Y$  using the Fourier transform, and  $N'$  denotes the length of the signal in the frequency domain. [Fig. 16](#) illustrates the feature space composed of  $\mu$  and  $\sigma$ , where data points corresponding to different penetration states are clearly distinguishable. The data from PP is nearly distinct, with only a minor overlap between the data from DP and FP. This suggests that data from a critical state may exhibit features inherent to two different

states, indicating that discerning critical states with precision using a straightforward feature space analysis is not feasible. Overall, the reconstructed signals exhibit good clustering in the feature space, both in the time and frequency domains. Consequently, the significant differences in the data across various penetration states are highly beneficial for the implementation of ML model classification.



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Fig. 16. Representation of the feature space for ML classification.

#### 4.5. Analysis of STN prediction results based on photoelectric signal

The computational station was configured with two Intel Xeon Gold 6246R processors, each running at 3.4GHz, each with 16 cores, and 256GB of DDR4 RAM operating at 4800MHz. Additionally, the station was equipped with an Nvidia RTX A5000 graphics card featuring 24GB of memory and 64-bit double precision. Parallel computing was facilitated by Nvidia Cuda 12.4, and computations were conducted on a 64-bit Windows 11 operating system. The coding was performed in Visual Studio Code, utilizing Python 3.9. [Table. 4](#) shows the hyperparameter combinations set during training for all models.

The hyperparameter combinations shown in Table. 5 were not optimized to test the benchmark performance of all models under the same criteria. Models were trained for 500 epochs with a batch size of 8 samples per iteration. The learning rate follows a cosine annealing schedule, varying between  $[1 \times 10^{-4}, 0]$ . The regularization function includes both layer normalization and dropout (with dropout rate of 0.1). In the fully connected layer, the Softmax function was used to compute the probability for each category.

Table 4. Hyper-parameters for all models.

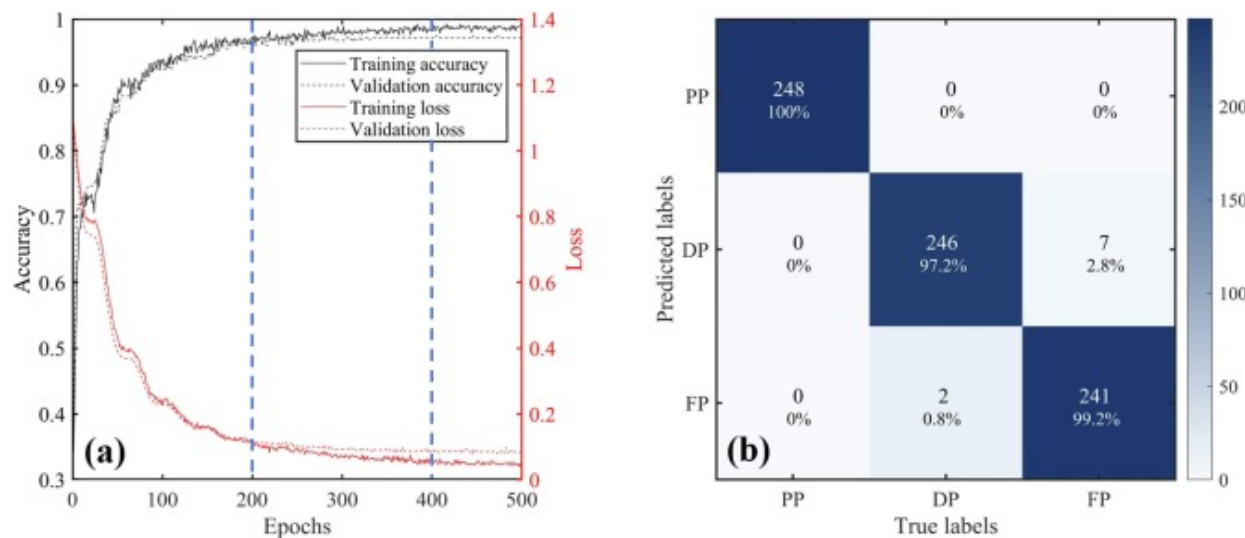
| Epochs | Batch size | Learning rate                               | Regularization                           | Optimizer | Activation function |
|--------|------------|---------------------------------------------|------------------------------------------|-----------|---------------------|
| 500    | 8          | Cosine annealing<br>$[1 \times 10^{-4}, 0]$ | Layer-normalization,<br>Dropout rate=0.1 | SGD       | Softmax             |

Table 5. Performance comparison of different DL models.

| Method          | Image size       | Model size<br>(#param.) | FLOPs | Test-set<br>accuracy | Training time<br>(minutes) | Throughput<br>(images/second) |
|-----------------|------------------|-------------------------|-------|----------------------|----------------------------|-------------------------------|
| STN             | 224 <sup>2</sup> | 27.5M                   | 4.27G | 98.8%                | 275.1                      | 1209.4                        |
| VGG19           | 224 <sup>2</sup> | 139.6M                  | 39.4G | 92.4%                | 351.3                      | 216.5                         |
| ResNet50        | 224 <sup>2</sup> | 23.5M                   | 8.2G  | 93.6%                | 319.2                      | 550.2                         |
| Inception-v3    | 299 <sup>2</sup> | 24.4M                   | 11.5G | 94.5%                | 337.5                      | 496.7                         |
| EfficientNet-B3 | 300 <sup>2</sup> | 10.7M                   | 1.9G  | 97.6%                | 239.8                      | 1387.2                        |

The dataset consists of 3720 images, divided into a training set, a validation set, and a test set following an 8:1:1 ratio. The training process of STN model is illustrated in Fig. 17(a). Notably, between 0 and 200 epochs, the model's RMSE value decreases rapidly, and after 400 epochs, the model essentially converges. The performance of the STN model on the

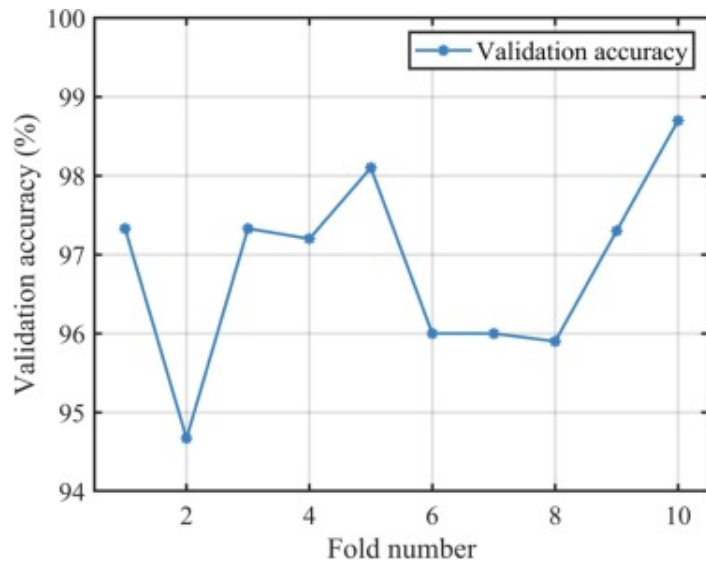
test set is depicted in Fig. 17(b), where the model achieves a 100% recognition accuracy for the PP category and a 96.4% accuracy for both the DP and FP categories. To verify that the STN model is not overfitted, 10-fold cross-validation was used. This process involved dividing the dataset into 10 equal parts, with the model trained on 9 parts and validated on the remaining part during each iteration. This procedure was repeated 10 times, each time using a different part as the validation set, ensuring that all samples were used for both training and validation. As shown in Fig. 18, the proposed model achieves no less than 94% accuracy in the validation process, with an average validation accuracy of 96.9%. This indicates that the model performs strongly across different data subsets, with no evidence of overfitting.



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Fig. 17. Training process and confusion matrix of STN: (a) training process; (b) confusion matrix for the test set.



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Fig. 18. 10-fold cross-validation accuracy of the STN.

To demonstrate the superiority of the STN model, it was compared with several models that utilize CNNs as their backbone networks. Given the impracticality of evaluating all possible CNN models, widely used and high-performance CNN architectures were selected for comparison: VGG19, ResNet50, Inception-v3, and EfficientNet [40], [41], [42], [43], [44]. Among these, VGG19, ResNet50, and Inception-v3 are classical CNN models that have been extensively validated and widely adopted. EfficientNet, a leading contemporary CNN model, achieved a top-1 accuracy of 84.3% on ImageNet, with fewer parameters and faster inference speed. Table 5 illustrates that EfficientNet exhibits the best prediction performance among the CNN models, with a recognition accuracy of 97.6%. In contrast, the STN model achieves a higher recognition accuracy of 98.8%.

To quantitatively assess the detailed actual performance of the proposed monitoring strategy, Precision (P), Recall (R), Specificity (S), and F1 score (F1) are employed as

evaluation metrics, with the calculation formulas provided as Eq. (17) through Eq. (20).

$$P = \frac{TP}{TP+FP} \quad (17)$$

$$R = \frac{TP}{TP+FN} \quad (18)$$

$$S = \frac{TN}{TN+FP} \quad (19)$$

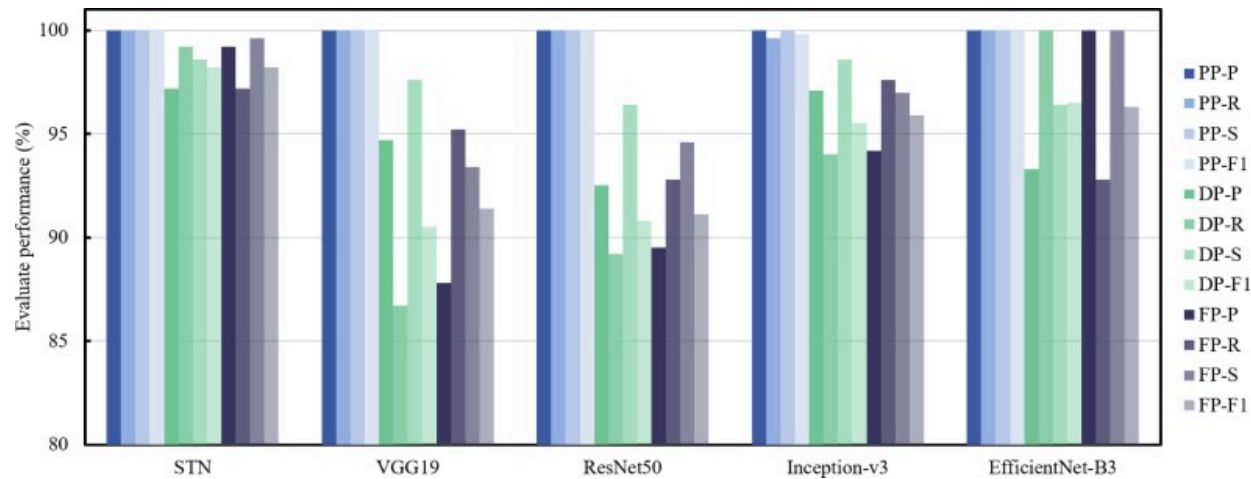
$$F1 = \frac{2 \times P \times R}{P+R} \quad (20)$$

where **TP**, **FN**, **TN**, and **FP** represent true positive, false negative, true negative, and false positive, respectively.

As shown in Fig. 19, the STN model effectively identifies the three penetration states.

Specifically, the proposed method achieves commendable scores across all four metrics: 98.8, 98.8, 99.4, and an average of 98.8 for the three categories. Compared to other CNN-based methods, STN demonstrates superior and more balanced performance overall.

Experimental results confirm the effectiveness of the proposed method for penetration state monitoring. However, EfficientNet-B3 outperforms STN in three metrics (DP-R, FP-P, and FP-S), highlighting the robust performance and generalization capabilities of CNNs, which should not be overlooked.



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Fig. 19. Evaluate performance for each category under different models.

#### 4.6. Potential limitations and challenges in industrial applications

While the proposed STN model has demonstrated impressive accuracy in identifying penetration states under controlled laboratory conditions, several challenges must be considered for industrial-scale deployment. First, variability in welding setups, such as differences in laser parameters, material properties, and environmental conditions, may affect the generalizability of the model. In real-world applications, welding systems may encounter variations in power settings, feed rates, or material thicknesses, which could impact the performance of the STN model if it is not sufficiently trained on diverse datasets. Additionally, materials used in industrial processes may exhibit different thermal properties, surface conditions, and thicknesses, further complicating the deployment. To address this, the model may require extensive retraining or fine-tuning when applied to new materials or setups, which can be resource-intensive. Another challenge is the scalability of the system. Real-time deployment would require sufficient

computational power and robust sensor integration to ensure the model can operate efficiently without causing delays in production processes.

#### 4.7. Ethical considerations

The introduction of AI and automation into welding processes raises concerns about potential job displacement, as tasks traditionally performed by skilled welders may increasingly be automated. However, rather than eliminating jobs, we believe that AI could shift the nature of work, creating new roles focused on the oversight, maintenance, and optimization of AI-driven systems. This transition will likely require workers to develop new skill sets, particularly in areas such as data analysis, machine learning, and the integration of AI with welding technologies. Hence, workforce retraining and education will be key to mitigating job displacement and ensuring that workers can adapt to the evolving demands of the industry. Additionally, the long-term effects of AI on the welding industry ecosystem are multifaceted. On one hand, AI could significantly improve productivity, quality control, and operational safety, making the welding process more efficient and reducing human exposure to hazardous environments. On the other hand, the widespread adoption of AI may create disparities between organizations that can afford to invest in such technologies and smaller businesses that may struggle to keep pace.

### 5. Conclusions

In this paper, an innovative online approach for identifying penetration states during pulsed laser welding of Al alloys is proposed. This method is based on the VMD-FEA and the STN model and can extract key features from multi-photoelectric signals to accurately identify penetration states. The correlation between photoelectric signal characteristics and penetration depth is examined from both mechanistic and data characterization perspectives. The main conclusions are as follows:

- (1) The time-dependent dynamics of the plasma plume and keyhole, captured through high-speed photography, exhibit a strong correlation with the photoelectric signal. Furthermore, when the laser power exceeds a threshold value of 3700W, both the plasma plume area and the photoelectric signal exhibit altered trends.
- (2) The proposed VMD-FEA algorithm not only enables the extraction of the signal component with the strongest correlation to penetration depth but also significantly improves the accuracy and efficiency compared to traditional methods. By decomposing and reconstructing the most relevant signal components, the VMD-FEA algorithm achieves a higher degree of correlation with penetration depth, leading to more precise monitoring and control of the welding process. Furthermore, the algorithm demonstrates superior clustering performance within the time–frequency domain feature space, ensuring that critical features are more effectively captured and analyzed.
- (3) In the proposed penetration state recognition model (STN), the STB backbone network, which takes the 2D time–frequency map of the photoelectric signal as input, can automatically extract features. Subsequently, a fully connected layer within the network identifies the penetration state, achieving a classification accuracy of 98.8%. Compared to other CNN-based models, the proposed method yielded superior results in terms of AR, PR, and F1.

## 6. Future works

Based on these findings, further research on the method's scalability and real-world application of the method is crucial. Real-time analysis of signal characteristics can help detect potential defects, such as porosity or uneven weld depth, enabling preemptive

corrective actions to ensure weld quality. Additionally, we aim to explore the integration of the method into adaptive control systems, developing simulations for real-time feedback loops that dynamically adjust laser power, welding speed, and oscillation frequency based on signal variations related to keyhole behavior and weld depth. Furthermore, it is essential to recognize the method's limitations in industrial settings. Photodetector signals are susceptible to environmental noise and the condition of the workpiece surface, both of which can affect monitoring precision. The effectiveness of the method is also highly dependent on specific process parameters and material properties, which may limit its applicability across various industrial scenarios. Finally, further optimization of system latency is necessary to guarantee real-time responsiveness in the dynamic context of production environments.

To address these challenges and expand the applicability of this method, future research will focus on the following directions:

- (1) Expanding experimental validation: We will be seeking additional industry partnerships to extend the experimental procedures to test the method under a wider variety of process parameters (such as laser power, welding speed, and focal length) and material conditions. This will enable a deeper understanding of how these variables affect the photoelectric signals and how the method can be optimized for use across a broader range of materials and welding scenarios.
- (2) Numerical simulations for signal analysis and control system design: To further explore the laser-material interaction mechanisms and their effects on optical radiation signals, future work will incorporate numerical simulations. These simulations will not only analyze how materials absorb and reflect laser energy but also serve as the basis for designing the feedback loops required for real-time adaptive control. By simulating the dynamic response of the welding system, we will evaluate how signal variations can trigger adjustments to laser power, speed, or other

parameters, providing a foundation for the integration of real-time control mechanisms into the welding process.

- (3) Multi-sensor approach: We will investigate integrating photoelectric sensing with optical coherence tomography (OCT) to enhance the method's real-time monitoring capabilities. By leveraging the complementary strengths of these technologies, this multi-sensor approach aims to capture both surface and subsurface phenomena, allowing for more accurate detection of dynamic changes in keyhole depth and internal defects during the welding process.
- (4) Model optimization and hybrid approaches: Future research will also focus on further optimizing the proposed models. We will explore hybrid models that integrate other deep learning architectures, such as Physics-Informed Neural Networks (PINNs), to incorporate domain-specific knowledge into the learning process. By embedding physical laws directly into the model, we aim to improve the robustness and generalization of the system, especially under varying welding conditions and materials.

## CRediT authorship contribution statement

**Da Zeng:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Di Wu:** Writing – review & editing, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hongxing Huang:** Writing – review & editing, Software, Methodology, Data curation. **Biao Peng:** Writing – review & editing, Visualization, Validation, Resources. **Yutong Wei:** Visualization, Validation, Software, Resources. **Hui Du:** Visualization, Software, Resources, Project administration. **Peilei Zhang:** Writing – review & editing, Validation, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Haichuan Shi:** Writing – review & editing, Visualization,

Validation, Methodology. **Qinghua Lu:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Conceptualization. **Xiaoyu Cai:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The data that has been used is confidential.

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